Casey's theorem and Fuss's theorem, formalised in Lean 4 with Mathlib

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Abstract

We report two small Lean 4 formalisations, built on Mathlib, of classical theorems about circles. The first is *Casey's theorem* (1866), the generalisation of Ptolemy's theorem in which the four points on a circle Γ are replaced by four circles tangent to Γ and distances are replaced by lengths of common tangents. We prove it in every real inner product space, for circles internally tangent to Γ and, through a signed radius, for any mixture of internally and externally tangent circles; with all radii zero it specialises to Mathlib's Ptolemy theorem, which the proof reuses. The second is *Fuss's theorem* (1797): if a quadrilateral is inscribed in a circle of radius R and its four side lines are tangent to a circle of radius r whose centre is at distance $d \neq R$ from the first centre, with $r > 0$, then $(R^2 - d^2)^2 = 2r^2(R^2 + d^2)$. We prove it in a Euclidean plane, together with the converse in the form of Poncelet's porism for quadrilaterals: under Fuss's relation every non-stalling four-step chain of the tangent construction (with the hypotheses of Theorem 9) closes, and if the inner circle lies inside the outer one, every point of the outer circle is a vertex of such a quadrilateral. Both original core developments are short (about 200 and 870 lines), contain no `sorry` and use only Lean's standard axioms. As far as we could determine, neither theorem had a formal proof before. The quadrilaterals need not be convex: convexity, and contact of the inner circle with the side segments, are not proved, so the textbook forms of the converse and of Poncelet closure for convex bicentric quadrilaterals are covered only in this narrower form. Version 2 adds Mathlib-only restatements with non-degenerate witnesses and a comparison with independently formalised textbook statements.

1 Introduction

This paper describes Lean 4 proofs of two classical theorems about configurations of circles. Casey's theorem is listed in Mathlib's "1000+ theorems" file, and Fuss's theorem is the case $n = 4$ of Poncelet's closure theorem, which is also listed; neither entry has a declaration recorded there [9].

Casey's theorem. Let Γ be a circle and let C_1, C_2, C_3, C_4 be circles internally tangent to Γ at points T_1, T_2, T_3, T_4 which lie on Γ in this cyclic order. For $i \neq j$ let t_{ij} be the length of a common external tangent of C_i and C_j; if C_i and C_j have centres at distance δ_{ij} and radii r_i, r_j, then $t_{ij} = \sqrt{\delta_{ij}^2 - (r_i - r_j)^2}$. Then

$$t_{12}\, t_{34} + t_{23}\, t_{41} = t_{13}\, t_{24}. \tag{1}$$

When all four circles shrink to points, $t_{ij} = |T_iT_j|$ and (1) is Ptolemy's theorem for the cyclic quadrilateral $T_1T_2T_3T_4$. The theorem was published by John Casey in 1866 [3] and appears in

his textbook [4]. There, and in most later accounts, it is stated together with a converse (four circles whose tangent lengths satisfy a relation of the form (1) are tangent to a common circle or line); we do not formalise the converse (Section 7.1 shows that a direct formalisation of it without non-degeneracy hypotheses is false). A standard variant allows some of the C_i to be externally tangent to Γ, with t_{ij} then the length of an *internal* common tangent when C_i and C_j touch Γ from different sides; our signed-radius form covers this.

Fuss's theorem. A quadrilateral is *bicentric* if it has a circumcircle Γ (centre O, radius R) and an incircle ω (centre I, radius r). With $d = |OI|$, Nicolaus Fuss showed in a paper presented in 1794 and printed in 1797 [6] that

$$(R^2 - d^2)^2 = 2r^2(R^2 + d^2), \qquad \text{equivalently} \qquad \frac{1}{(R-d)^2} + \frac{1}{(R+d)^2} = \frac{1}{r^2}. \tag{2}$$

The triangle analogue is the Euler–Chapple formula $d^2 = R(R - 2r)$. Poncelet's closure theorem (1822) [10, 5] explains both: given two circles, one inside the other, start at a point of the outer circle, draw a tangent to the inner circle, take its second intersection with the outer circle, draw the other tangent from there, and repeat. Whether the resulting chain closes after n steps does not depend on the starting point. For $n = 3$ the closure condition is the Euler–Chapple relation and for $n = 4$ it is Fuss's relation (2). The triangle case, including the Euler–Chapple relation as a closure criterion, is formalised in a companion paper [2]; here we treat $n = 4$.

Contributions.

- A proof of Casey's theorem (Theorem 3) in an arbitrary real inner product space, for four circles internally tangent to a circle Γ, and a signed-radius version (Theorem 4) that allows circles externally tangent to Γ. The only nontrivial input is Mathlib's Ptolemy theorem; the new work is the tangent-length formula $t_{ij} = |T_iT_j|\sqrt{R - r_i}\sqrt{R - r_j}/R$ (Lemma 7).
- A proof of Fuss's theorem (Theorem 8) in a Euclidean plane for quadrilaterals that need not be convex, with the single non-degeneracy hypothesis $d \neq R$; closure of the tangent construction after four steps under (2) (Theorem 9); existence of a bicentric quadrilateral through every point of Γ when ω lies inside Γ (Theorem 10); and the resulting "all or nothing" statements (Corollary 11).
- Lemmas that tie the definitions to their informal reading (Section 5), including a formal non-vacuity theorem for Fuss's theorem and a counterexample showing that the hypothesis $d \neq R$ cannot be dropped.
- (Version 2.) Restatements of the main theorems with Mathlib-only types, each with an explicit non-degenerate witness, and a comparison with textbook statements formalised independently by an AI model (Section 7.1). The forward textbook Fuss statement is covered with the additional hypothesis $d < R$ and without the square-characterisation clause; its converse and the Poncelet statements are covered only with side-line tangency, without proofs of convexity or contact on the side segments.

Availability. The two developments are separate Lake projects, published at `https://github.com/jpbald93/casey` and `https://github.com/jpbald93/fuss`. Line references in this paper are to commits `5f6252c` (Casey) and `671efdc` (Fuss); the Casey references are also valid at the pinned commit `a83c43b`, which adds `Casey/Bridge.lean`

and leaves `Basic.lean` and `Main.lean` unchanged. Version 2 of this paper adds, in a new commit of each repository (Casey `8e9058891620926763871d85a204a7a980354f21`, Fuss `c5a5915b223e5f7fce56e5c5dcaba450ef36ad48`), the Layer-14 and Layer-15 files `Casey/L14.lean` (174 lines), `Casey/L15Classical.lean` (394), `Fuss/L14.lean` (354) and `Fuss/L15Classical.lean` (418) of Section 7.1; that commit changes only the root import files, `gate.sh` and the README, and all earlier statements and library files are unchanged. Both use Lean `v4.33.1` and Mathlib revision `0df444a3` (tag `v4.33.1`), pinned in `lake-manifest.json`; this is the same Mathlib as in the companion paper [2]. The original Casey core consists of `Casey/Basic.lean` and `Casey/Main.lean` (200 lines, plus its original 2-line root file). The original Fuss core consists of six files under `Fuss/` (866 lines): `Plane.lean`, `Forward.lean`, `Porism.lean`, `Steps.lean`, `Geometry.lean` and `Example.lean`. Two lemmas of `Fuss/Geometry.lean` are copied from the triangle development `poncelet-triangles` of [2], with the namespace changed and a provenance note added to the docstring (Section 2.2). Supporting material (the Layer-14 and Layer-15 reports, the blind writer's statements and report, the transcript audit, and the reviews of this version with the fixes made in response) is in the folder `Casey-Fuss` of the repository `https://github.com/jpbald93/LEAN-verification`.

Table 1: Main formal statements. Links point to the declaration line at the commit listed under Availability.

Lean name	In this paper	Source
Repo `casey`		
`Casey.tangentCentre`	Def. 1	`Basic.lean`:28
`Casey.tangentLength`	Def. 1	`Basic.lean`:33
`Casey.tangentCentre_zero`	Section 5.1	`Basic.lean`:43
`Casey.tangentLength_zero`	Section 5.1	`Basic.lean`:47
`Casey.dist_tangentCentre_center`	Lemma 15	`Basic.lean`:51
`Casey.dist_tangentCentre_sq_sub_sq`	Lemma 6	`Basic.lean`:62
`Casey.tangentLength_tangentCentre`	Lemma 7	`Basic.lean`:77
`Casey.cospherical_of_dist_eq`	Section 3.2	`Main.lean`:25
`Casey.casey_of_le`	Theorem 4	`Main.lean`:34
`Casey.casey`	Theorem 3	`Main.lean`:61
`Casey.tangentCentre_spec`	Lemma 15	`Main.lean`:76
`Casey.tangentCentre_spec_external`	Lemma 15	`Main.lean`:90
`Casey.casey_zero_radii_eq_ptolemy`	Corollary 5	`Main.lean`:102
Repo `fuss`		
`Fuss.parallel_of_inner_eq_zero`	Lemma 13	`Plane.lean`:22
`Fuss.eq_or_eq_or_eq_of_inner`	Lemma 13	`Plane.lean`:51
`Fuss.vertex_smul`	Lemma 13	`Plane.lean`:83
`Fuss.vertex_rel`	Lemma 14	`Forward.lean`:21
`Fuss.not_three_on_line`	Section 4.4	`Forward.lean`:57
`Fuss.fuss_vec`	Section 4.4	`Forward.lean`:71
`Fuss.vertex_unique`	Section 4.5	`Porism.lean`:22
`Fuss.chain_distinct`	Section 4.6	`Porism.lean`:31
`Fuss.closes_vec`	Section 4.5	`Porism.lean`:59
`Fuss.first_tangent`	Section 4.6	`Steps.lean`:42
`Fuss.other_tangent`	Section 4.6	`Steps.lean`:64
`Fuss.second_point`	Section 4.6	`Steps.lean`:87
`Fuss.step`	Section 4.6	`Steps.lean`:119
`Fuss.isTangent_of_mem_orthRadius`	Section 2.2	`Geometry.lean`:33
`Fuss.IsBicentric`	Def. 2	`Geometry.lean`:96

Table 1, continued

Lean name	In this paper	Source
`Fuss.fuss_of_isBicentric`	Theorem 8	`Geometry.lean`:105
`Fuss.poncelet_chain_closes`	Theorem 9	`Geometry.lean`:142
`Fuss.exists_quad_vec`	Section 4.6	`Geometry.lean`:175
`Fuss.exists_bicentric_of_fuss`	Theorem 10	`Geometry.lean`:202
`Fuss.fuss_iff_exists`	Corollary 11	`Geometry.lean`:241
`Fuss.fuss_iff_forall`	Corollary 11	`Geometry.lean`:257
`Fuss.exists_iff_forall`	Corollary 11	`Geometry.lean`:273
`Fuss.fuss_iff_reciprocal`	Proposition 12	`Geometry.lean`:280
`Fuss.nonvacuous`	Proposition 16	`Example.lean`:18

2 Setting and representation

Throughout, V is a real inner product space and P a metric affine space over it:

```
[NormedAddCommGroup V] [InnerProductSpace ℝ V] [MetricSpace P] [NormedAddTorsor V P]
```

For Casey's theorem no dimension assumption is made. For Fuss's theorem we add `[Fact (Module.finrank ℝ V = 2)]`, so that P is a Euclidean plane; the model plane `EuclideanSpace ℝ (Fin 2)` is used for the non-vacuity example. We write $\angle a\,p\,c$ for Mathlib's unoriented angle `EuclideanGeometry.angle`.

2.1 Casey: circles as centre and signed radius

In the Casey development a circle is not a Mathlib `EuclideanGeometry.Sphere`; it is given by its centre and a real number, and tangency is built into the choice of centre.

Definition 1. Let Γ have centre O and radius R, and let T be a point.

- `Casey.tangentCentre` $O\ T\ R\ r = O + (1 - r/R)(T - O)$ is the centre of the circle of radius r internally tangent to Γ at T.
- `Casey.tangentLength` $c_1\ c_2\ r_1\ r_2 = \sqrt{|c_1c_2|^2 - (r_1 - r_2)^2}$, with Mathlib's `Real.sqrt` (which returns 0 on negative arguments).

For $0 \le r < R$ and $T \in \Gamma$ the point $c =$ `tangentCentre` $O\ T\ R\ r$ lies on the segment OT at distance $R - r$ from O and at distance r from T, so the circle with centre c and radius r touches Γ from inside at T (Lemma 15). For $r < 0$ the same formula puts c on the ray OT beyond T, at distance $R + |r|$ from O and $|r|$ from T: the circle of radius $|r|$ with centre c touches Γ from outside at T. So one real parameter $r \le R$ encodes both kinds of tangency, and we call it a *signed radius*. With signed radii r_i, r_j, the quantity $|c_ic_j|^2 - (r_i - r_j)^2$ is the squared length of the common external tangent when r_i, r_j have the same sign, and of the common internal tangent (for which the relevant radius sum is $|r_i| + |r_j| = r_i - r_j$) when $r_i \ge 0 \ge r_j$. This is the classical convention for the mixed form of Casey's theorem.

Cyclic order. Following Mathlib's Ptolemy theorem, the cyclic order of $T_1, \dots, T_4$ on Γ is expressed by a point p with $\angle T_1\,p\,T_3 = \pi$ and $\angle T_2\,p\,T_4 = \pi$: the diagonals T_1T_3 and T_2T_4 meet at a point strictly inside both. For four *distinct* points on a circle this is the usual meaning

of "T_1, T_2, T_3, T_4 in this cyclic order" (up to reversal); each angle hypothesis is strict betweenness (`EuclideanGeometry.angle_eq_pi_iff_sbtw`). The hypotheses also admit repeated adjacent points, e.g. $T_1 = T_2$ and $T_3 = T_4$ at opposite ends of a diameter; the identities remain true there. Mathlib's theorem is

```
theorem mul_dist_add_mul_dist_eq_mul_dist_of_cospherical {a b c d p : P}
    (h : Cospherical ({a, b, c, d} : Set P)) (hapc : ∠ a p c = π) (hbpd : ∠ b p d = π) :
    dist a b * dist c d + dist b c * dist d a = dist a c * dist b d
```

(`EuclideanGeometry.mul_dist_add_mul_dist_eq_mul_dist_of_cospherical`, by Candales and Davidson), and our statements take the same two angle hypotheses.

2.2 Fuss: Mathlib spheres and tangent lines

In the Fuss development the circles are Mathlib spheres `Γ ω : Sphere P`, with $R = \Gamma$.radius, $r = \omega$.radius and $d =$ `dist Γ.center ω.center`. The formal statements are written with these expressions; we abbreviate them as R, r, d. A line is `line[ℝ, A, B]`, the affine span of $\{A, B\}$. Tangency is Mathlib's `EuclideanGeometry.Sphere.IsTangent`: an affine subspace is tangent to ω if it contains a point $T \in \omega$ and lies in the tangent space ω.`orthRadius` T, the affine subspace through T orthogonal to the radius at T.

Definition 2. `Fuss.IsBicentric` $\Gamma\ \omega\ A\ B\ C\ D$ holds if $A, B, C, D \in \Gamma$ are pairwise distinct and the four side lines AB, BC, CD, DA are tangent to ω:

```
def IsBicentric (Γ ω : Sphere P) (A B C D : P) : Prop :=
  A ∈ Γ ∧ B ∈ Γ ∧ C ∈ Γ ∧ D ∈ Γ ∧
    A ≠ B ∧ B ≠ C ∧ C ≠ D ∧ D ≠ A ∧ A ≠ C ∧ B ≠ D ∧
    ω.IsTangent line[ℝ, A, B] ∧ ω.IsTangent line[ℝ, B, C] ∧ ω.IsTangent line[ℝ, C, D]
    ∧
    ω.IsTangent line[ℝ, D, A]
```

This definition does not ask for convexity, nor that the points of tangency lie on the side segments. It therefore includes crossed (self-intersecting) quadrilaterals whose four side lines touch ω. Section 5 discusses this choice.

The proofs work in vectors relative to the centre I of ω. A point X becomes $x = X - I$; the centre of Γ becomes $o = O - I$, so $\|o\| = d$; membership $X \in \Gamma$ becomes $\|x - o\|^2 = R^2$, and for $T \in \omega$ the condition $X \in \omega$.`orthRadius` T becomes $\langle x, t \rangle = r^2$ (`Fuss.mem_orthRadius_iff`). The translation lemmas `Fuss.vec_of_mem`, `Fuss.mem_of_vec` and `Fuss.vec_of_mem_center` are one-line rewrites. In the other direction, `Fuss.isTangent_of_mem_orthRadius` says that in a plane, for $T \neq$ the centre of ω, the line through two distinct points of the tangent space at T is tangent to ω; it and `Fuss.exists_mem_sphere` (a circle of positive radius in a nontrivial space has a point) are the two lemmas copied from [2].

3 Casey's theorem

3.1 Statements

Theorem 3 (Casey)**.** *Let $R > 0$, let $T_1, \dots, T_4$ be points with $|T_i O| = R$, let p be a point with $\angle T_1\, p\, T_3 = \pi$ and $\angle T_2\, p\, T_4 = \pi$, and let $0 \le r_i < R$. Put $c_i =$* `tangentCentre` *$O\ T_i\ R\ r_i$ and $t_{ij} =$* `tangentLength` *$c_i\ c_j\ r_i\ r_j$. Then $t_{12}t_{34} + t_{23}t_{41} = t_{13}t_{24}$.*

```
theorem casey {O T₁ T₂ T₃ T₄ p : P} {R r₁ r₂ r₃ r₄ : ℝ} (hR : 0 < R)
    (hr₁ : 0 ≤ r₁ ∧ r₁ < R) (hr₂ : 0 ≤ r₂ ∧ r₂ < R) (hr₃ : 0 ≤ r₃ ∧ r₃ < R)
    (hr₄ : 0 ≤ r₄ ∧ r₄ < R)
    (h₁ : dist T₁ O = R) (h₂ : dist T₂ O = R) (h₃ : dist T₃ O = R) (h₄ : dist T₄ O = R)
    (h₁₃ : ∠ T₁ p T₃ = π) (h₂₄ : ∠ T₂ p T₄ = π) :
    tangentLength (tangentCentre O T₁ R r₁) (tangentCentre O T₂ R r₂) r₁ r₂ *
        tangentLength (tangentCentre O T₃ R r₃) (tangentCentre O T₄ R r₄) r₃ r₄ +
      tangentLength (tangentCentre O T₂ R r₂) (tangentCentre O T₃ R r₃) r₂ r₃ *
        tangentLength (tangentCentre O T₄ R r₄) (tangentCentre O T₁ R r₁) r₄ r₁ =
      tangentLength (tangentCentre O T₁ R r₁) (tangentCentre O T₃ R r₃) r₁ r₃ *
        tangentLength (tangentCentre O T₂ R r₂) (tangentCentre O T₄ R r₄) r₂ r₄
```

Theorem 4 (Casey, signed radii)**.** *The conclusion of Theorem 3 holds under the same hypotheses with $0 \le r_i < R$ replaced by $r_i \le R$ (*`Casey.casey_of_le`*). In particular it holds for any combination of circles internally tangent ($0 \le r_i < R$) and externally tangent ($r_i < 0$) to Γ.*

Theorem 3 is proved from Theorem 4 in one line. The value $r_i = R$ is allowed in Theorem 4; then $c_i = O$ and the "circle" is Γ itself, every t_{ij} involving i is 0 by Lemma 7, and the identity reduces to $0 = 0$.

Corollary 5 (Ptolemy)**.** *Under the hypotheses on R, T_i and p of Theorem 3, $|T_1T_2|\,|T_3T_4| + |T_2T_3|\,|T_4T_1| = |T_1T_3|\,|T_2T_4|$ (*`Casey.casey_zero_radii_eq_ptolemy`*).*

This is obtained by specialising `Casey.casey` to $r_1 = \cdots = r_4 = 0$ and simplifying with `Casey.tangentCentre_zero` ($c_i = T_i$) and `Casey.tangentLength_zero` ($t_{ij} = |T_iT_j|$). It is of course weaker than the Mathlib theorem used in the proof; its purpose is to show that the formal statement contains Ptolemy's theorem.

3.2 Proof

The proof follows the classical one and has three steps.

Lemma 6 (key identity)**.** *If $R \neq 0$ and $|T_1O| = |T_2O| = R$, then for all real r_1, r_2, with $c_i =$* `tangentCentre` *O T_i R r_i,*

$$|c_1c_2|^2 - (r_1 - r_2)^2 = \frac{|T_1T_2|^2\,(R - r_1)(R - r_2)}{R^2}.$$

Proof sketch. Put $u = T_1 - O$, $v = T_2 - O$, $a = 1 - r_1/R$ and $b = 1 - r_2/R$, so $c_1 - c_2 = au - bv$ and $\|u\| = \|v\| = R$. Expand $\|au - bv\|^2$ and $\|u - v\|^2$ with `norm_sub_sq_real`, substitute the norms, clear denominators (`field_simp`) and close with `ring`. No hypothesis on the signs of r_i or on the dimension is used. □

Lemma 7 (tangent length)**.** *If $R > 0$, $r_1, r_2 \le R$ and $|T_1O| = |T_2O| = R$, then*

$$t_{12} = \texttt{tangentLength}\ c_1\ c_2\ r_1\ r_2 = \frac{|T_1T_2|\,\sqrt{R - r_1}\,\sqrt{R - r_2}}{R}.$$

Proof sketch. By Lemma 6 the argument of the square root is the square of the right-hand side, which is non-negative because $R - r_i \ge 0$ and $R > 0$; so `Real.sqrt_sq` applies. □

The lemma is where the hypothesis $r_i \le R$ enters: it makes $R - r_i \ge 0$, so that $\sqrt{R-r_1}\sqrt{R-r_2} = \sqrt{(R-r_1)(R-r_2)}$ and the argument of the outer square root is non-negative. It is also the only place where the truncation of `Real.sqrt` could matter, and the lemma shows that it does not.

Proof of Theorem 4. The four points are cospherical (`Casey.cospherical_of_dist_eq`, with centre O and radius R), so Mathlib's Ptolemy theorem gives $|T_1T_2||T_3T_4| + |T_2T_3||T_4T_1| = |T_1T_3||T_2T_4|$. Rewrite each of the six tangent lengths with Lemma 7. Writing $s_i = \sqrt{R - r_i}$, the left-hand side of (1) becomes $(|T_1T_2||T_3T_4| + |T_2T_3||T_4T_1|)\, s_1s_2s_3s_4/R^2$ (an auxiliary identity proved by `field_simp`), and the right-hand side becomes $|T_1T_3||T_2T_4|\, s_1s_2s_3s_4/R^2$. Ptolemy's identity finishes the proof. □

The original Casey core is two files and 200 lines. The main work was choosing a representation in which Lemma 6 is a polynomial identity: the circles are parametrised by their points of tangency and signed radii, not by arbitrary centres subject to a tangency hypothesis.

4 Fuss's theorem and Poncelet's porism for quadrilaterals

4.1 Statements

Theorem 8 (Fuss)**.** *Let P be a Euclidean plane, Γ, ω circles with $r > 0$ and $d \ne R$. If* `IsBicentric` $\Gamma\ \omega\ A\ B\ C\ D$*, then $(R^2 - d^2)^2 = 2r^2(R^2 + d^2)$.*

```
theorem fuss_of_isBicentric [Fact (Module.finrank ℝ V = 2)] (Γ ω : Sphere P) {A B C D :
    P}
    (hr : 0 < ω.radius) (hd : dist Γ.center ω.center ≠ Γ.radius)
    (h : IsBicentric Γ ω A B C D) :
    (Γ.radius ^ 2 - dist Γ.center ω.center ^ 2) ^ 2 =
      2 * ω.radius ^ 2 * (Γ.radius ^ 2 + dist Γ.center ω.center ^ 2)
```

Theorem 9 (closure)**.** *Let P be a Euclidean plane, $r > 0$, $d \ne R$, and assume Fuss's relation* (2)*. Let $A_0, \dots, A_4 \in \Gamma$ and $T_0, \dots, T_4 \in \omega$ be such that A_i lies on the tangents to ω at T_i and T_{i+1} ($i = 0, \dots, 3$) and A_4 lies on the tangent at T_4, with $T_i \ne T_{i+1}$ and $A_i \ne A_{i+1}$ for $i = 0, \dots, 3$. Then $A_4 = A_0$ and $T_4 = T_0$ (*`Fuss.poncelet_chain_closes`*).*

In words: start at A_0 on the tangent at T_0; take the *other* tangent from A_0 (touching at $T_1 \ne T_0$) and its *other* intersection $A_1 \ne A_0$ with Γ; repeat. After four steps the construction is back where it started. "A lies on the tangent at T" is written $A \in \omega$.`orthRadius` T with $T \in \omega$. The theorem quantifies over all such chains, so it is independent of how the choices in the construction are made.

Theorem 10 (existence, porism)**.** *Let P be a Euclidean plane, $r > 0$ and $d + r < R$ (so ω lies strictly inside Γ), and assume* (2)*. Then for every $A \in \Gamma$ there are B, C, D with* `IsBicentric` $\Gamma\ \omega\ A\ B\ C\ D$ (`Fuss.exists_bicentric_of_fuss`)*.*

Corollary 11 (all or nothing)**.** *Let P be a Euclidean plane, $r > 0$ and $d + r < R$. The following are equivalent: (i) some bicentric quadrilateral for Γ, ω exists; (ii) every point of Γ is a vertex of one; (iii) Fuss's relation* (2) *holds. The three equivalences (i)⇔(iii), (ii)⇔(iii) and (i)⇔(ii) are* `Fuss.fuss_iff_exists`*,* `Fuss.fuss_iff_forall` *and* `Fuss.exists_iff_forall`*.*

Proposition 12. *For $0 \le d < R$ and $r \ne 0$, $(R^2 - d^2)^2 = 2r^2(R^2 + d^2)$ if and only if $1/(R-d)^2 + 1/(R+d)^2 = 1/r^2$ (*`Fuss.fuss_iff_reciprocal`*).*

4.2 Three facts about the plane

The vector proofs of the forward relation and of closure use dimension two through the following lemma and its consequences; the existence construction (`Fuss.first_tangent`) and the affine tangency translation (`Fuss.isTangent_of_mem_orthRadius`) also use dimension two directly.

Lemma 13. *Let* $\dim V = 2$.

(a) If $e \neq 0$ *and* $\langle a, e\rangle = \langle b, e\rangle = 0$, *then* $\|a\|^2 b = \langle a, b\rangle a$ *(*`Fuss.parallel_of_inner_eq_zero`*).*

(b) A line meets a circle in at most two points: if $\langle a - b, n\rangle = \langle a - d, n\rangle = 0$ *with* $n \neq 0$ *and* b, d *are at the same distance from* c *as* a, *then two of* a, b, d *coincide (*`Fuss.eq_or_eq_or_eq_of_inner`*).*

(c) If $\|x\|^2 = \|y\|^2 = r^2$, $x \neq y$ *and* $\langle v, x\rangle = \langle v, y\rangle = r^2$ *(so* v *is the intersection of the tangents to the circle* $\|\cdot\| = r$ *at* x *and* y*), then* $(r^2 + \langle x, y\rangle)\, v = r^2(x + y)$ *(*`Fuss.vertex_smul`*), and* $r^2 + \langle x, y\rangle \neq 0$ *if* $r \neq 0$ *(*`Fuss.vertex_denom_ne`*).*

Part (a) is proved with Mathlib's area form on an oriented two-dimensional space (`Orientation.inner_mul_areaForm_sub`). Further consequences are that a point lies on at most two tangents of a circle (`Fuss.tangent_points`) and that a tangent line meets a circle in at most two points (`Fuss.line_points`).

4.3 The key identity

Place I at the origin and put $o = O - I$ and $k = R^2 - \|o\|^2 = R^2 - d^2$. For a tangency point x of ω define the *shifted point* $y = kx + 2r^2 o$, and let

$$F = k^2 - 2r^2(R^2 + \|o\|^2),$$

so that Fuss's relation is $F = 0$.

Lemma 14 (shifted tangency points)**.** *Let* $\dim V = 2$ *and* $r \neq 0$. *If* $\|x\|^2 = \|y\|^2 = r^2$, $x \neq y$, $\langle a, x\rangle = \langle a, y\rangle = r^2$ *and* $\|a - o\|^2 = R^2$ *(the tangents at* x *and* y *meet at a point* a *of* Γ*), then*

$$\langle kx + 2r^2 o,\ ky + 2r^2 o\rangle = -r^2 F.$$

This is `Fuss.vertex_rel`.

Proof sketch. Write $s = r^2 + \langle x, y\rangle$, which is nonzero by Lemma 13(c). From $sa = r^2(x + y)$ one gets $s^2\|a\|^2 = 2r^4 s$ (take the squared norm) and $s\langle a, o\rangle = r^2(\langle x, o\rangle + \langle y, o\rangle)$ (take the inner product with o). Expanding the left-hand side of the claim and $\|a - o\|^2 = R^2$, the product of s with (left side $+ r^2 F$) is an explicit polynomial linear combination of these three equations, certified by `linear_combination`; cancelling $s \neq 0$ gives the claim. □

The certificate was also checked independently with SymPy (`code/verify_lc.py`, Section 6). The point of the lemma is that the right-hand side depends only on R, r, d: the shifted tangency points of the two sides through any vertex have the same inner product $\mu = -r^2 F$.

4.4 Fuss's theorem

The affine statement is reduced to the vector statement `Fuss.fuss_vec`: the tangency points $T_1, \dots, T_4$ are extracted from the four `IsTangent` hypotheses, both endpoints of each side lie in the tangent space at its tangency point (the field `le_orthRadius` of `EuclideanGeometry.Sphere.IsTangentAt`), and $d \neq R$ becomes $k \neq 0$ (using $R \geq 0$, which follows from $A \in \Gamma$).

Proof of `fuss_vec`. Let the vertices be a, b, c, e and the tangency points of ab, bc, ce, ea be $x_1, \dots, x_4$. *Distinctness.* Three distinct points of Γ cannot lie on one tangent line (`Fuss.not_three_on_line`, from Lemma 13(b)). Since the vertices are pairwise distinct, this gives $x_i \neq x_{i+1}$ (indices mod 4) and also $x_1 \neq x_3$, $x_2 \neq x_4$. *Equal inner products.* By Lemma 14 at the four vertices, the shifted points $y_i = kx_i + 2r^2 o$ satisfy $\langle y_4, y_1\rangle = \langle y_1, y_2\rangle = \langle y_2, y_3\rangle = \langle y_3, y_4\rangle = \mu$. Since $k \neq 0$, $x \mapsto kx + 2r^2 o$ is injective, so $y_1 \neq y_3$ and $y_2 \neq y_4$. *Contradiction if $F \neq 0$.* Suppose $F \neq 0$, so $\mu \neq 0$. Then y_2 and y_4 are both orthogonal to $y_1 - y_3 \neq 0$, so by Lemma 13(a), $\|y_4\|^2 y_2 = \langle y_4, y_2\rangle y_4$. Taking the inner product with y_3 gives $\|y_4\|^2\mu = \langle y_4, y_2\rangle\mu$, hence $\|y_4\|^2 = \langle y_4, y_2\rangle$ and so $\|y_4\|^2(y_2 - y_4) = 0$. Also $y_4 \neq 0$ since $\langle y_3, y_4\rangle = \mu \neq 0$. Hence $y_2 = y_4$ and $x_2 = x_4$, a contradiction. So $F = 0$. □

No hypothesis places ω inside Γ or I inside the quadrilateral. This differs from the triangle case, where the analogous argument gives Euler's relation only when ω is the incircle and a different relation, $d^2 = R(R + 2r)$, when it is an excircle [2]; for quadrilaterals there is no second branch.

4.5 Closure

Proof of Theorem 9 (via `closes_vec`). Now $F = 0$, so by Lemma 14 consecutive shifted points are orthogonal: $\langle y_i, y_{i+1}\rangle = 0$ for $i = 0, \dots, 3$. Each y_i is nonzero: from $kx = -2r^2 o$ and $\|x\| = r$ one gets $k^2 = 4r^2\|o\|^2$, and together with $F = 0$ this forces $r^2 k = 0$, which is excluded. Also $\|y_i - 2r^2 o\|^2 = k^2 r^2$ for all i, so all y_i lie on one circle centred at $2r^2 o$. *Step 1.* y_1 and y_3 are both orthogonal to $y_2 \neq 0$, so they are parallel (Lemma 13(a)); since $y_0 \perp y_1$ and $y_1 \neq 0$, also $y_0 \perp y_3$. *Step 2.* Now y_0, y_2, y_4 are all orthogonal to y_3, so they lie on one line; they also lie on the circle above. By Lemma 13(b), two of them coincide. If $x_0 = x_2$, then a_0 and a_1 would both lie on the two distinct tangents at x_1 and x_2, forcing $a_0 = a_1$ by `Fuss.vertex_unique`; so $x_0 \neq x_2$, and similarly $x_2 \neq x_4$. Hence $x_4 = x_0$. *Step 3.* The points a_0, a_3, a_4 of Γ all lie on the tangent at $x_4 = x_0$, which meets Γ in at most two points (`Fuss.line_points`). The case $a_3 = a_4$ is excluded by hypothesis; the case $a_0 = a_3$ would put a_0 on three distinct tangents (at x_0, x_1 and x_3), which leads to a contradiction with `Fuss.tangent_points` and `Fuss.vertex_unique`. Hence $a_4 = a_0$. □

4.6 Existence

Theorem 9 assumes that a chain is given. To prove Theorem 10 one has to construct one, and this is where $d + r < R$ is used.

- Every point of Γ lies strictly outside ω, and every point of ω strictly inside Γ (`Fuss.outside_of_mem`, `Fuss.inside_of_mem`).
- From a point outside ω there is a tangent (`Fuss.first_tangent`); the tangency point is written down explicitly with Mathlib's right-angle rotation `Orientation.rightAngleRotation`.
- Given one tangency point x from a, its reflection in the line $0a$ is a second, different one (`Fuss.other_tangent`).

- The line through $a \in \Gamma$ and a point x strictly inside Γ meets Γ again at an explicit point $b \neq a$, which lies on the same tangent if x is its tangency point (`Fuss.second_point`).

`Fuss.step` combines the last two items into one step of the construction. `Fuss.exists_quad_vec` applies it four times from a first tangent at the given point a, closes the chain with `Fuss.closes_vec`, and obtains the two diagonal inequalities $a_0 \neq a_2$, $a_1 \neq a_3$ from `Fuss.chain_distinct`; these do not need Fuss's relation. Finally `Fuss.exists_bicentric_of_fuss` translates back to P and proves the four tangencies with `Fuss.isTangent_of_mem_orthRadius`.

Corollary 11 follows: for (i)⇒(iii) and (ii)⇒(iii) use Theorem 8, noting that $d + r < R$ and $r > 0$ imply $d \neq R$; for (iii)⇒(ii) use Theorem 10, and for (ii)⇒(i) the fact that Γ has a point (`Fuss.exists_mem_sphere`). Proposition 12 is a direct computation.

5 Correctness of the formal statements

The Lean kernel checks the proofs, but not that the statements say what the informal theorems say. We explain each hypothesis and give lemmas that support the intended reading.

5.1 Casey

Lemma 15. *Let $R > 0$, $|TO| = R$ and $c =$* `tangentCentre` *$O\ T\ R\ r$.*

(a) If $0 \leq r < R$, then $|cO| = R - r$ and $|Tc| = r$ (`Casey.tangentCentre_spec`*).*

(b) If $r \leq 0$, then $|cO| = R + (-r)$ and $|Tc| = -r$ (`Casey.tangentCentre_spec_external`*).*

For all r, $|cO| = |R - r|$ (`Casey.dist_tangentCentre_center`*).*

Two circles with centres at distance $R - r$ and radii $R > r$ are internally tangent, and T is a common point; with distance $R + |r|$ they are externally tangent. So (a) and (b) say that `tangentCentre` places the circle correctly. Conversely, every circle of radius $r < R$ internally tangent to Γ at T has its centre on the segment OT at distance $R - r$ from O, which is the point `tangentCentre`. This uniqueness statement is formalised in the Layer-14 file for Mathlib's internal tangency `EuclideanGeometry.Sphere.IsIntTangentAt` (`Casey.l14_center_of_isIntTangentAt`, Section 7.1).

The hypotheses. `hR : 0 < R` makes Γ a genuine circle; $\mathtt{h}_i$ put T_i on Γ; $\mathtt{hr}_i$ restrict to circles inside Γ, $r_i = 0$ being allowed and meaning the point T_i; and the angle hypotheses encode cyclic order exactly as in Mathlib's Ptolemy theorem. Without cyclic order the statement is false: Ptolemy's identity fails for the order (T_1, T_3, T_2, T_4), and so does Casey's (this is the negative control in Section 6). The strict inequality $r_i < R$ in Theorem 3 excludes the degenerate "circle" Γ itself; Theorem 4 shows that it is not needed for the identity.

Non-vacuity. By Corollary 5 the hypotheses of Theorem 3 are those of Ptolemy's theorem plus the radius bounds, and $r_i = 0$ satisfies the bounds. Ptolemy's hypotheses are satisfied, for example, by the vertices of a square inscribed in Γ with $p = O$. This point-circle instance is not a Lean theorem, but version 2 adds a non-degenerate witness with four distinct circles of positive radius (`Casey.l14w_casey`, Section 7.1).

Bridge lemmas. The module `Casey/Bridge.lean` connects the two definitions to the classical notions. `Casey.tangentCentre_mem_line` and `Casey.collinear_tangentCentre` put c on the line OT, so O, c and T are collinear; `Casey.tangentCentre_tangent` gives, for every r, $|TO| = R$, $|Tc| = |r|$ and $|cO| = |R - r|$, the internal and external tangency distances. `Casey.tangentLength_sq` is $t^2 = d^2 - (r_1 - r_2)^2$ when the right side is nonnegative, and `Casey.tangentLength_eq_dist_of_perp` shows that for a point and a circle the formula is the length $|pX|$ of a tangent segment whose radius cX is perpendicular to pX, with $|pX|^2 = |pc|^2 - r^2$.

What is not claimed. The tangent length is defined by the formula, not as the length of a segment of a common tangent line; that the two agree is the Pythagorean theorem in the right trapezoid formed by the centres and the points of tangency, and is not formalised. The theorem is stated for four circles tangent to a circle, not to a line.

5.2 Fuss

The hypotheses of Theorem 8. `hr : 0 < ω.radius` excludes a point-circle. `hd : d ≠ R` excludes the case where the centre I of ω lies on Γ, and it is needed: with $d = R$ the relation reads $0 = 2r^2 \cdot 2R^2$, which is false for $r > 0$, but bicentric quadrilaterals in the sense of Definition 2 still exist. The script `code/check_fuss.py` constructs one with $R = d = 2$, $r = 1/2$, $O = (2, 0)$ and $I = (0, 0)$, with vertices in the order $A \approx (0.0645, 0.5039)$, $B \approx (3.8755, 0.6946)$, $C \approx (0.0645, -0.5039)$, $D \approx (3.8755, -0.6946)$. Exactly: with tangency vectors $x_1 = (3/20, \sqrt{91}/20)$, $x_2 = (-1/40, \sqrt{399}/40)$, $x_3 = (3/20, -\sqrt{91}/20)$, $x_4 = (-1/40, -\sqrt{399}/40)$, the vertices are $A_i = (x_i + x_{i+1})/(4(1/4 + \langle x_i, x_{i+1}\rangle))$ (indices mod 4). They are pairwise distinct, lie on Γ, and the four side lines are tangent to ω. In this configuration ω is not inside Γ and the quadrilateral is crossed. The theorem does *not* assume that ω is inside Γ, that I is inside the quadrilateral, or that the quadrilateral is convex; so it applies to crossed quadrilaterals whose side lines touch ω, and Fuss's relation holds for these too. The distinctness conditions in Definition 2, including the diagonal ones $A \neq C$ and $B \neq D$, are all used in the proof (to get $x_1 \neq x_3$ and $x_2 \neq x_4$).

The hypotheses of Theorems 9 and 10. In Theorem 9 the conditions $T_i \neq T_{i+1}$ and $A_i \neq A_{i+1}$ express "take the other tangent" and "take the other intersection". Without them the chain could stall, for instance with $A_1 = A_0$. The theorem does not require ω to be inside Γ, but outside that case a chain may fail to exist (from a point inside ω there is no tangent). Theorem 10 requires $d + r < R$ because the construction steps need it; this is also the classical setting of Poncelet's porism.

Proposition 16 (non-vacuity, `Fuss.nonvacuous`)**.** *In* `EuclideanSpace` $\mathbb{R}$ *(`Fin 2`) there exist circles Γ, ω with $r > 0$, $d + r < R$ and distinct centres such that Fuss's relation holds and every point of Γ is a vertex of a bicentric quadrilateral. The proof uses the witnesses Γ with centre $(1, 0)$ and radius 7, and ω with centre $(0, 0)$ and radius $24/5$ (so $d + r = 29/5 < 7$).*

Here $d = 1$ and $(49 - 1)^2 = 2304 = 2 \cdot (24/5)^2 \cdot 50$; in reciprocal form, $1/6^2 + 1/8^2 = 25/576 = 1/(24/5)^2$. So the hypotheses of Theorem 10 are satisfiable with non-concentric circles, and `IsBicentric` is not an empty notion. (In the concentric case $d = 0$, Fuss's relation is $R = \sqrt{2}\, r$: the square.)

What is not claimed. Definition 2 asserts tangency of the side *lines*. It does not say that ω is the incircle of a convex quadrilateral $ABCD$. When ω is inside Γ the quadrilateral produced by the

construction is classically convex, with the tangency points on the sides, but we have not proved this, so Theorem 10 does not assert it. Accordingly, the Layer-15 comparison (Section 7.1) covers the textbook converse of Fuss's theorem and the textbook Poncelet closure statements only with `Fuss.IsBicentric` in place of a convex quadrilateral whose contact points lie on the side segments.

6 Numerical checks

Both repositories contain Python scripts in `code/` that use the same definitions as the Lean code. They are sanity checks of the statements, not part of the proofs.

Casey (`code/check_casey.py`). Centres and tangent lengths are computed with the formulas of Definition 1, and four points in cyclic order are sampled on random circles, with a fixed seed.

- 20 000 configurations with $0 \le r_i \le 0.999R$ (about 20% with one radius set to 0): worst relative residual of (1) $7.5 \cdot 10^{-12}$.
- 20 000 configurations with signed radii $-2R \le r_i \le 0.999R$: worst relative residual $5.2 \cdot 10^{-11}$.
- *Negative control:* 20 000 further configurations, relabelled in the non-cyclic order (T_1, T_3, T_2, T_4), violate (1) by more than 10^{-6} in all 20 000 cases (smallest relative residual $1.1 \cdot 10^{-5}$).
- All radii 0 (Ptolemy): residual 0.

Fuss (`code/check_fuss.py`, `code/verify_lc.py`).

- For 2 000 random R, $d < 0.9R$ and r chosen to satisfy (2), the tangent construction from a random point closes after four steps; maximum error $3.6 \cdot 10^{-13}$.
- *Negative control:* for $R = 3$, $d = 1$ and r perturbed by $\pm 10\%$ from the Fuss value, the smallest closure gap is approximately 3.13.
- A polynomial shift identity related to Lemma 14 (with shift point $-2r^2o/k$) holds on 100 random inputs, and the non-vacuity example gives $2304 = 2304$.
- Independently of the construction, 294 configurations were found by starting from four tangency points on ω and solving for concyclic vertices (not necessarily convex); all satisfy (2), with maximum relative defect $3.5 \cdot 10^{-13}$.
- The $d = R$ counterexample of Section 5: four distinct vertices, all at distance 2 from O, while (2) reads $0 = 4$.
- `code/verify_lc.py` checks with SymPy the polynomial `linear_combination` certificates in `Fuss.vertex_rel`, in `Fuss.first_tangent` and `Fuss.other_tangent`, and in `Fuss.fuss_iff_reciprocal`; all seven residuals expand to 0.

The outputs are stored in `code/check_fuss.out` and `code/verify_lc.out`. The Casey script and `check_fuss.py` were re-run while this paper was prepared and reproduced the stored figures (the current `check_fuss.py` additionally prints the vertices of the $d = R$ example).

7 Verification

Both projects were built with prebuilt Mathlib on a separate build machine; the machine that prepared this paper did not run Lean. According to the development notes, each was rebuilt from an empty build directory (`rm -rf .lake/build; lake build`): Casey completed with 2 454 jobs and Fuss with 8 713 jobs, each with exit status 0, no errors and no warnings.

A grep of the library sources (`Casey/`, `Casey.lean`, `Fuss/`, `Fuss.lean`) for `sorry`, `admit`, `native_decide`, `axiom`, `#eval`, `run_cmd`, `set_option`, `macro`, `elab`, `syntax`, `notation` and `import Lean` finds nothing; we repeated it on the copies of the sources used for this paper.

`#print axioms` was run from a scratch file outside the library (`scratch/Ax.lean`); the output (`scratch/ax.out`) is, for Casey,

```
'Casey.dist_tangentCentre_sq_sub_sq' depends on axioms: [propext, Classical.choice, Quot.sound]
'Casey.tangentLength_tangentCentre' depends on axioms: [propext, Classical.choice, Quot.sound]
'Casey.casey_of_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'Casey.casey' depends on axioms: [propext, Classical.choice, Quot.sound]
'Casey.casey_zero_radii_eq_ptolemy' depends on axioms: [propext, Classical.choice, Quot.sound]
'Casey.tangentCentre_spec' depends on axioms: [propext, Classical.choice, Quot.sound]
'Casey.tangentCentre_spec_external' depends on axioms: [propext, Classical.choice, Quot.sound]
```

and for Fuss,

```
'Fuss.fuss_of_isBicentric' depends on axioms: [propext, Classical.choice, Quot.sound]
'Fuss.poncelet_chain_closes' depends on axioms: [propext, Classical.choice, Quot.sound]
'Fuss.exists_bicentric_of_fuss' depends on axioms: [propext, Classical.choice, Quot.sound]
'Fuss.fuss_iff_exists' depends on axioms: [propext, Classical.choice, Quot.sound]
'Fuss.fuss_iff_forall' depends on axioms: [propext, Classical.choice, Quot.sound]
'Fuss.exists_iff_forall' depends on axioms: [propext, Classical.choice, Quot.sound]
'Fuss.fuss_iff_reciprocal' depends on axioms: [propext, Classical.choice, Quot.sound]
'Fuss.nonvacuous' depends on axioms: [propext, Classical.choice, Quot.sound]
'Fuss.fuss_vec' depends on axioms: [propext, Classical.choice, Quot.sound]
'Fuss.closes_vec' depends on axioms: [propext, Classical.choice, Quot.sound]
```

These are Lean's three standard axioms. Every theorem in Table 1 either has its own report or is used in the proof of a theorem that has one; the definitions do not need reports.

The author's name gate `namecheck.py` (not included in the verification repository) checks that every Lean identifier cited in the text exists as a declaration in the two source trees or in the pinned Mathlib, and that every source link lands on a declaration line (for table rows, of the declaration named in that row). Each repository also has a `gate.sh` that rejects forbidden tokens, builds the library, generates its own `#print axioms` queries for every declaration cited in this paper (18 for Casey, 34 for Fuss) and fails on any axiom beyond the standard three. On the published commits both print `PASS`. In the version-2 commit the gates also check the Layer-14 and Layer-15 declarations (38 for Casey, 84 for Fuss) and again print `PASS`. `tests/tamper.sh` plants six kinds of defect in scratch copies (a `sorry`, a split `axiom`, a fake `#eval` report, a macro, a weakened statement, a wrapped extra axiom); in both repositories the gate rejects all six, each with a nonzero exit status and the expected failure message.

The step that matters most, comparing the formal statements with the informal theorems, cannot be automated; Section 5 is meant to help with it. The author has checked each statement in Table 1 against its informal counterpart by hand.

7.1 Layers 14 and 15

Version 2 adds two further checks, in new files of the published repositories: `Casey/L14.lean`, `Casey/L15Classical.lean`, `Fuss/L14.lean` and `Fuss/L15Classical.lean`. The Layer-15 files

use the namespaces `Casey.Classical` and `Fuss.Classical`. The earlier library files are byte-identical. The gates check 38 (Casey) and 84 (Fuss) declarations, all with the standard axioms only. `tests/tamper.sh` rejects all six planted defects in both repositories.

Layer 14: statements in Mathlib's language and non-vacuity. Nine main results are restated with Mathlib-only types and proved from the library theorems. `Casey.l14_casey` restates Theorem 3. Its circles are Mathlib spheres C_i with C_i`.IsIntTangentAt` Γ T_i, and the tangent lengths are written out as $\sqrt{|c_ic_j|^2-(r_i-r_j)^2}$, because Mathlib has no tangent-length notion. No extra hypothesis is needed, and the statement also allows $r_i = R$. `Casey.l14_casey_ptolemy` restates Corollary 5. The bridge `Casey.l14_center_of_isIntTangentAt` identifies the centre with `Casey.tangentCentre`. In the Fuss file, `Fuss.l14_fuss`, `Fuss.l14_poncelet_chain_closes`, `Fuss.l14_exists_bicentric`, `Fuss.l14_fuss_iff_exists`, `Fuss.l14_fuss_iff_forall`, `Fuss.l14_exists_iff_forall` and `Fuss.l14_fuss_iff_reciprocal` restate Theorems 8–10, Corollary 11 and Proposition 12, with `Fuss.IsBicentric` unfolded. A mechanical check of the elaborated types found no constant of the two libraries.

Witness theorems prove every hypothesis on an explicit non-degenerate configuration:

- `Casey.l14w_casey`: Γ of radius 5 and four pairwise distinct circles of radii $1, 2, 3, 4$ tangent at the pairwise distinct points $\pm 5, \pm 5i$. All six radicands are positive.
- `Fuss.l14w_fuss`, `Fuss.l14w_poncelet_chain_closes` and `Fuss.l14w_exists_bicentric`: $\Gamma = ((1,0),7)$, $\omega = ((0,0),24/5)$ and an explicit kite with pairwise distinct vertices and non-concentric circles.
- `Fuss.l14w_fuss_iff`: both sides of the equivalences hold for (Γ,ω). For a second inner circle of radius 1, Fuss's relation fails and no bicentric quadrilateral exists.
- `Fuss.l14w_fuss_iff_reciprocal`: $(R,d,r) = (7,1,24/5)$ satisfies both sides, and $r = 1$ neither.

Proposition 16 is already a Mathlib-only statement and is not restated. The externally tangent (signed-radius) reading of Theorem 4 is not restated, because a Mathlib form would need mixed tangent lengths $\sqrt{\delta^2-(r_i \pm r_j)^2}$. No restatement turned out false or needed an extra hypothesis. The kite is convex, with its contact points on the sides, but this is not proved in Lean.

Layer 15: comparison with textbook statements. Another agent collected textbook statements of Casey's theorem (with the signed-radius form and the converse), of Fuss's theorem and its converse, and of Poncelet's closure theorem for quadrilaterals, with citations (Casey, Johnson, Dörrie, Yiu, Poncelet, Bos et al.). The classical sources take a bicentric quadrilateral to be convex. An AI model was given only these statements; it saw neither the code nor this paper, and its transcript was audited (a transcript audit in the verification repository; the collector's transcript passed the same audit, and the collector recorded editions and pages for each source). It wrote nine Lean statements, two of them radicand obligations, with its own definitions on `EuclideanSpace` $\mathbb{R}$ `(Fin 2)`. Among those definitions are a convex quadrilateral whose incircle touches every side segment, tangency as a unique common point, and cyclic order as crossing chords. A comparer then proved each statement, copied verbatim, from the libraries, or classified it. The result: 4 covered exactly, 4 narrower, 1 false as written. No library file was changed.

- *Covered*: Casey's theorem for disjoint internally tangent circles, the signed-radius form with $t_{ij} = |K_iK_j|\sqrt{(R-\rho_i)(R-\rho_j)}/R$, and both radicand obligations (`Casey.Classical.cov_`

`CaseyFuss1_a`, `Casey.Classical.cov_CaseyFuss1_b` and the two `_obligation` theorems). The disjointness hypothesis is not needed.

- *Narrower*, Fuss's theorem (`Fuss.Classical.cov_CaseyFuss2_a_part`): proved for the writer's convex quadrilaterals, with the reciprocal and solved forms and $\sqrt{2}\,r \le R$ with equality iff $d = 0$. It needs the extra hypothesis $d < R$, which is classically automatic but not derived (the repository's notion allows $d = R$). The clause "$d = 0$ iff the quadrilateral is a square" is not formalised.

- *Narrower*, converse of Fuss (`Fuss.Classical.cov_CaseyFuss2_b_part`): the quadrilateral produced is bicentric only in the sense of Definition 2. Convexity and contact points on the side segments are not proved.

- *Narrower*, Poncelet closure (`Fuss.Classical.cov_CaseyFuss3_a_part`): from one convex bicentric quadrilateral, a quadrilateral through every point of Γ is obtained, the writer's tangent chain closes after exactly four steps, and there are infinitely many such quadrilaterals. All of these quadrilaterals are bicentric only in the weaker sense: convexity and contact on segments are not proved.

- *Narrower*, "existence iff Fuss's relation" (`Fuss.Classical.cov_CaseyFuss3_b_part`): the forward direction is proved as written. The converse direction and the "through every point" clause hold only with `Fuss.IsBicentric`.

- *Writer statement false as written*: the converse of Casey's theorem (`Casey.Classical.not_classical_CaseyFuss1_c`). With C_0 the unit circle, C_1 the point $(1, 0)$ and $C_2 = C_3$ the point $(-1, 0)$, all tangent lengths from C_0 vanish, so the relation holds, but no circle or line touches all four. The textbook converse tacitly assumes distinct, non-degenerate circles. The paper does not claim the converse.

New proofs in the Layer-15 files: for Casey, `Casey.Classical.tc_eq` (a centre at the right distances is `Casey.tangentCentre`), `Casey.Classical.int_dist` and `Casey.Classical.ext_dist` (the writer's internal and external tangency give the centre distances), `Casey.Classical.signed_bridge`, `Casey.Classical.no_mid`, `Casey.Classical.cyc_angles` (crossing chords give the angle-π cyclic order), `Casey.Classical.casey_signed`, `Casey.Classical.radicand_nonneg` and the counterexample above. For Fuss: `Fuss.Classical.tangentAt_of_side` (tangency to a side segment gives Mathlib tangency), `Fuss.Classical.ne_of_not_collinear`, `Fuss.Classical.isBicentric_of_with` (the writer's convex bicentric quadrilateral is `Fuss.IsBicentric`), `Fuss.Classical.fuss_of_with`, `Fuss.Classical.fuss_algebra`, `Fuss.Classical.fuss_of_recip`, `Fuss.Classical.tangentAt_of_lineTangent`, `Fuss.Classical.not_collinear_of_mem`, `Fuss.Classical.not_three`, and the infinitude of Γ.

8 Related work

Our searches, on 30 September 2026, covered the pinned Mathlib source and its `docs/100.yaml` and `docs/1000.yaml`, the Lean community's list of formalised theorems, GitHub code, repository and pull-request search, the Isabelle Archive of Formal Proofs (through its GitHub mirror), HOL Light, and web search. They are not exhaustive, and the claims below are "as far as we could determine". A search of the Lean Zulip chat for "Casey", "Fuss" and "bicentric" on 30 September 2026 returned no relevant results.

Casey's theorem. Mathlib has no declaration mentioning Casey. `docs/1000.yaml` has an entry `Q1370316` ("Casey's theorem") with no declaration or author recorded [9], and the theorem is not in `docs/100.yaml`; the Lean community's list of the 1000+ theorems shows no formalisation. GitHub code search for "Casey", "Casey theorem" and "casey_theorem" in Lean files returned no hits; "generalized Ptolemy" returned only prose datasets and an unrelated notion in SnapPy; searches of Mathlib pull requests for "Casey" and of repositories for "casey theorem" found nothing. The Isabelle AFP has an entry for Ptolemy's theorem [1] but none for Casey's, and HOL Light has `100/ptolemy.ml` but nothing on Casey [7]. Our proof reuses Mathlib's Ptolemy theorem [8] unchanged.

Fuss's theorem and Poncelet's porism. Mathlib has nothing on Fuss's theorem or bicentric polygons, and `docs/1000.yaml` lists "Poncelet's closure theorem" (`Q1785610`) with no declaration [9]. GitHub code search for "Fuss" and "bicentric" in Lean files found only unrelated uses (comments, and Fuss–Catalan numbers), and Mathlib pull-request search for "poncelet" and "bicentric" found nothing. Web search found no formalisation of Fuss's theorem in the Isabelle AFP, HOL Light, Coq/Rocq or Mizar.

The closest prior work is Wang's Lean 4 repository `wwylele/Poncelet` [11] (Lean `v4.34.0`, September 2026). Its theorem `EuclideanGeometry.poncelet` is the general Poncelet porism for n-gons and a circle inside a circle: if one proper n-gon is inscribed in the outer circle and circumscribed about the inner one, then one exists through every point of the outer circle. This is more general than our Theorem 10 in n, but it contains no metric closure condition, and in particular not Fuss's relation for $n = 4$ (nor our Theorem 8). According to its README, its main theorem depends on `native_decide`, and hence on an axiom beyond the standard three. We read the repository but did not build it. The companion paper [2] treats $n = 3$, with the Euler–Chapple relation as closure criterion, by a different method (explicit tangent constructions and Euler's formula). The vector method used here, based on the shifted tangency points of Lemma 14, was not reused from it; only two general lemmas were copied (Section 2.2).

9 Limitations and further work

- *Casey: converse.* The converse (four circles whose tangent lengths satisfy a Casey relation are tangent to a common circle or line) is substantially harder and was not attempted. The Layer-15 writer's formalisation of it is false as written: it lacks non-degeneracy hypotheses (`Casey.Classical.not_classical_CaseyFuss1_c`).

- *Casey: degenerate and mixed cases.* Circles tangent to a line (the limit $R \to \infty$) are not treated. The signed radius covers circles externally tangent to Γ, with t_{ij} the internal tangent between an internally and an externally tangent circle; variants that use the internal tangent between two circles on the same side of Γ are not treated.

- *Casey: cyclic order.* Cyclic order is expressed only through Mathlib's diagonal-intersection hypothesis; there is no version with oriented angles or with `Sbtw`.

- *Casey: representation.* In the main files circles are given by `tangentCentre`, not as Mathlib spheres with a tangency hypothesis. The Layer-14 file restates Theorem 3 with Mathlib spheres and `EuclideanGeometry.Sphere.IsIntTangentAt`, proves that every internally tangent circle has the `tangentCentre` form, and gives a non-degenerate witness; the externally tangent (signed-radius) case of Theorem 4 is not restated there (Section 7.1). The identification of `tangentLength` with the length of a tangent segment is not formalised (Section 5.1).

- *Fuss: convexity.* Theorem 10 produces a quadrilateral in the sense of Definition 2; that it is convex, with ω as its incircle and the tangency points on the sides, is classically true when ω is inside Γ but is not proved. For the same reason the textbook converse and Poncelet statements of Layer 15 are proved only in a narrower form, and the forward textbook statement needs the extra hypothesis $d < R$ (Section 7.1).
- *Fuss: range of the converse.* Closure (Theorem 9) needs only $d \neq R$, but existence needs $d + r < R$.
- *Fuss: packaging.* The construction is stated with explicit tangency points (`orthRadius`), not with Mathlib's `EuclideanGeometry.Sphere.tangentsFrom` and `EuclideanGeometry.Sphere.secondInter` as in the triangle development [2].
- *Fuss: n-gons.* Only $n = 4$ is treated. The shifted-point argument is specific to quadrilaterals; closure conditions for larger n (Fuss and Steiner gave several) would need new ideas.
- *Duplication.* Two lemmas are copied from the triangle development; a shared library or upstreaming to Mathlib would remove this.

Acknowledgements and use of AI

The formal proofs, the numerical scripts and a draft of this paper were prepared with the help of a large-language-model AI assistant. The author compared each formal statement in Table 1 with its informal counterpart by hand. Every proof is checked by the Lean kernel, and the author takes responsibility for all content. The Layer-14 and Layer-15 files of version 2 (Section 7.1) were also written by AI agents, the Layer-15 textbook statements by a writer that saw neither the code nor this paper.

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